# Initial Purchase Request

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REV A / 09 OCT 2026
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## 1.0 PURPOSE

This is the team's first consolidated purchase request. Procurement lead time is the main schedule risk this season, so everything the next two months need is requested now, in two tiers. **Tier A** (33 lines) is test supplies for test cards TC-01 to TC-04. Buy them locally this week: the TC-01 and TC-02 results feed the 13 Oct design decisions. **Tier B** (40 lines) goes to the university's formal process.

**Budget holds.** 12 Tier B lines depend on the aircraft sizing due at the 13 Oct decision gate (motors, ESCs, propellers, packs, servos, tubes, reel actuators). For these, the category, quantity and a reference price are known now, so the budget can be reserved today. Part numbers follow on 14 Oct.

**Prices.** 25 lines carry a vendor's listed price, checked on 9 Oct 2026 and listed in Section 7.0. Together they cover 68 % of the value. Every other price is a planning estimate and needs a quote.

REQUEST AT A GLANCE

| TIER A, LOCAL | TIER B, BUY NOW |
|---|---|
| 1,280.20 USD | 3,713.20 USD |
| BUDGET HOLDS | TOTAL |
| 3,850.96 USD | 8,844.36 USD |
| TOTAL | LINES |
| 2,749 KWD | 73 |

## 2.0 HOW TO READ THIS REQUEST

Each table covers one category. Every line gives the quantity, unit price and extended price in US dollars, plus two status words:

- **BUY**: the specification is fixed; order now.
- **HOLD**: reserve the budget now; the exact part follows the sizing gate.
- **listing**: the vendor's listed price on 9 Oct 2026. It is a reference, not a quote.
- **estimate**: a planning figure that has not been verified.

Prices exclude shipping, duties and VAT. Dinar figures use 0.3108 KWD per US dollar, and the one sterling listing uses 1.3229 USD per pound (market rates, 8 Oct 2026). The spreadsheet `PR-01_initial_request.xlsx` holds the same lines with vendor links, need-by dates and KWD columns.

The need-by dates are proposals:

- TC-01 and TC-02 supplies: 11 Oct.
- Other Tier A lines: 14 Oct.
- Tier B buy-now lines: 1 Nov.
- Budget holds: ordered 14 Oct, needed by 30 Nov.
- Travel crate: 1 Mar 2027.

## 3.0 NOT IN THIS REQUEST

These costs belong in the proposal budget but are paid outside this request:

- Fly-off registration fee: US$550 per team (§2.6).
- AIAA student membership: US$37 per member per year (aiaa.org, checked 9 Oct). Every team member needs one (§1.1, §2.1).
- AMA membership for the pilot (§1.2): price to confirm with the AMA.
- Travel to Tucson and aircraft freight: a separate travel request.
- Contingency: set by the team lead.

## 4.0 PURCHASING CONSTRAINTS FROM THE RULES

These come from the 2026–27 contest rules (September 2026 draft). A substitution that breaks one of them can fail tech inspection, so please check with the team before swapping a part on these lines.

- **Radio.** Use ExpressLRS 2.4 GHz receivers with PWM outputs (team decision, 8 Oct 2026). CRSF-only receivers need a flight controller, and flight controllers are banned (§3.2.1, §4.2.1). Buy FCC versions: the radio must use a legal US frequency (§3.2.1).
- **Batteries.** Every pack (propulsion, receiver and sensor) must be unaltered, commercial off-the-shelf, and carry the manufacturer's label stating chemistry, capacity, voltage and C-rating (§3.2.3, §3.1.1).
- **Propulsion packs.** No pack may exceed 100 Wh, and neither may the total on the airplane (§3.2.3.2). Keep the original packaging: tech inspection checks it (§4.2.3). Lithium packs ship as dangerous goods, so buy them locally if possible.
- **Charging sacks.** Commercial, unaltered and manufacturer-labelled. Every battery is stored and charged in one (§3.2.3).
- **Fuses.** Blade fuses in a separate harness. The rating may not exceed the pack's maximum continuous current or 100 A (§3.2.3.1, §3.2.3.2).
- **Arming plug.** Its wire must match the pack leads' gauge or be heavier (§3.2.2). The reference pack has 10 AWG leads.
- **ESCs.** The BEC must be disabled, and a separate receiver battery is mandatory (§3.2.3, FAQ A.3).
- **Motors, propellers, electronics.** Use unmodified commercial motors and commercial propellers. The airplane may carry no camera, GPS, autopilot or flight controller (§3.2.1).
- **Latches.** Positive mechanical latches only; magnets are not acceptable (§4.2.1).

## 5.0 TIER A: LOCAL PURCHASE THIS WEEK

These lines support test cards TC-01 to TC-04 and the electronics bench. Where a petty-cash or reimbursed local purchase is allowed, it beats the formal process for these lines. Tier A total: US$1,280.20.

TABLE 01 Measurement and safety. 6 lines, subtotal US$151.99.

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| A01 | **Precision bench scale**<br>3 kg capacity, 0.1 g resolution, tare. Weighs bodies, ballast and containers to 0.01 oz (0.28 g).<br>Ref. TC-01, TC-02, TC-04. Example model: Duratool D03412 (3 kg × 0.1 g). | 2 | 15.00 | 30.00 | BUY<br>estimate |
| A02 | **Digital hanging scale**<br>50 kg × 10 g with hold function. Tension check; weighing payload bundles and airframes.<br>Ref. TC-01, TC-04. | 1 | 15.00 | 15.00 | BUY<br>estimate |
| A03 | **Vane anemometer**<br>Handheld, 0.3–30 m/s, reads m/s and ft/min.<br>Ref. TC-01. Upper limit 30 m/s (98 ft/s). | 1 | 26.99 | 26.99 | BUY<br>listing |
| A04 | **Digital caliper**<br>150 mm, 0.01 mm. Foam thickness and permanent set; simulator size match (±1/8 in). Skip if the lab has one.<br>Ref. TC-02, §3.1.3. | 1 | 20.00 | 20.00 | BUY<br>estimate |

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| A05 | **Safety glasses**<br>Impact-rated (ANSI Z87.1 or EN 166).<br>Ref. TC-01, TC-02. | 6 | 4.00 | 24.00 | BUY<br>estimate |
| A06 | **Earmuffs**<br>Passive hearing protection for blower runs.<br>Ref. TC-01. | 3 | 12.00 | 36.00 | BUY<br>estimate |
| | **Subtotal** | | | **151.99** | |

TABLE 02 TC-01 blower rig. 6 lines, subtotal US$456.00.

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| A07 | **Leaf blower**<br>Highest air volume available locally; target ≥ 15 m³/min (530 cfm). 230 V / 50 Hz or cordless; variable speed preferred.<br>Ref. TC-01. US reference model. Buy the local equivalent and check its air-volume rating. | 1 | 359.00 | 359.00 | BUY<br>listing |
| A08 | **Duct and adapters**<br>Rigid round duct, 100 mm and 150 mm × 1 m; reducer to the blower nozzle; aluminium tape; clamps.<br>Ref. TC-01. | 1 | 40.00 | 40.00 | BUY<br>estimate |
| A09 | **Drinking straws**<br>Plain straight straws, about 6–8 mm × 200 mm, for the flow straightener.<br>Ref. TC-01. About 500 straws. | 1 | 10.00 | 10.00 | BUY<br>estimate |
| A10 | **Ball-bearing swivels**<br>Fishing ball-bearing swivels, ≥ 50 lb rated.<br>Ref. TC-01. A swivel lets any roll tendency show as rotation. | 10 | 1.20 | 12.00 | BUY<br>estimate |
| A11 | **Braided PE line**<br>50–80 lb braided PE (Dyneema-type) fishing line, ≥ 150 yd.<br>Ref. TC-01, TC-02, tow-line candidate. | 1 | 20.00 | 20.00 | BUY<br>estimate |
| A12 | **Ballast assortment**<br>Steel nuts and washers M5–M10 and lead-free sinkers, about 1 kg.<br>Ref. TC-01, TC-02. Ballast must be internal to the sensor (§3.1.1). | 1 | 15.00 | 15.00 | BUY<br>estimate |
| | **Subtotal** | | | **456.00** | |

TABLE 03 TC-02 drop test. 6 lines, subtotal US$244.95.

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| A13 | **Foam sheet assortment**<br>At least 4 types: EPE (2 densities), EPS (2 densities), XPS, open-cell PU, EVA; 10, 20, 25 and 50 mm; about 0.5 m² per type and thickness.<br>Ref. TC-02, Script 5. Ask each supplier for density and the compressive stress–strain curve (crush stress, densification strain). | 1 | 120.00 | 120.00 | BUY<br>estimate |
| A14 | **Plywood and MDF sheets**<br>3 mm and 6 mm, laser-cutter bed size.<br>Ref. TC-02 containers and fixture. | 6 | 8.00 | 48.00 | BUY<br>estimate |
| A15 | **Brass rod**<br>1/4 in (6.35 mm) brass, 300 mm long, for the quick-release pin replica. 6 mm metric is acceptable for testing.<br>Ref. TC-02. The contest pin is 1/4 in brass (§3.1.3). | 2 | 6.00 | 12.00 | BUY<br>estimate |
| A16 | **Welded steel rings**<br>Welded (not split) stainless rings, wire diameter ≤ 3.0 mm, inside diameter ≥ 13 mm.<br>Ref. TC-02, TC-04. Rule: ID ≥ 1/2 in (12.7 mm), cross-section ≤ 1/8 in (3.175 mm) (§3.1.3). | 10 | 1.00 | 10.00 | BUY<br>estimate |

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| A17 | **Rigging hardware assortment**<br>Mini stainless snap hooks, quick links, ball-lock quick-release pins, crimp sleeves.<br>Ref. TC-02, TC-04, SR-12. | 1 | 30.00 | 30.00 | BUY<br>estimate |
| A18 | **High-g accelerometer (optional)**<br>Adafruit H3LIS331 ±400 g breakout (product 4627), logged by a Pico (A24).<br>Ref. TC-02 peak g. Optional. Without it, TC-02 still runs on damage checks and foam set. | 1 | 24.95 | 24.95 | BUY<br>listing |
| | **Subtotal** | | | **244.95** | |

TABLE 04 TC-03 lights. 5 lines, subtotal US$112.26.

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| A19 | **Colour LED stars**<br>Cree XP-E2 single-emitter 20 mm star: red, red-orange, amber (1000 mA max).<br>Ref. TC-03. | 6 | 5.71 | 34.26 | BUY<br>listing |
| A20 | **White LED stars**<br>Cree XP-E2 or XP-G white, 20 mm star.<br>Ref. TC-03. | 4 | 6.00 | 24.00 | BUY<br>estimate |
| A21 | **LED optics**<br>20 mm star optics, narrow (about 15°) and medium (about 30°).<br>Ref. TC-03. | 8 | 2.00 | 16.00 | BUY<br>estimate |
| A22 | **Constant-current LED drivers**<br>350–700 mA buck modules with a PWM dimming input.<br>Ref. TC-03. | 6 | 3.00 | 18.00 | BUY<br>estimate |
| A23 | **5 mm LEDs and LED strip**<br>Ultra-bright 5 mm assortment (white, red, amber, green); 5 V SMD strip, 1 m each of white and red.<br>Ref. TC-03. | 1 | 20.00 | 20.00 | BUY<br>estimate |
| | **Subtotal** | | | **112.26** | |

TABLE 05 EE bench. 5 lines, subtotal US$110.00.

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| A24 | **Microcontroller boards**<br>Raspberry Pi Pico (RP2040): TC-03 flash-pattern driver, test-box prototype, A18 logger.<br>Ref. TC-03, EE lane. | 4 | 4.00 | 16.00 | BUY<br>listing |
| A25 | **1S LiPo packs, labelled**<br>COTS 1S LiPo, 300–1000 mAh; the label shows chemistry, capacity, voltage and C-rating.<br>Ref. TC-03, sensor battery. The sensor battery must meet §3.2.3 (§3.1.1). | 4 | 8.00 | 32.00 | BUY<br>estimate |
| A26 | **1S LiPo charger**<br>USB charger for the A25 packs.<br>Ref. TC-03. | 1 | 12.00 | 12.00 | BUY<br>estimate |
| A27 | **Inline current meter**<br>USB / DC power meter, 0–5 A, 3–30 V.<br>Ref. TC-03. | 1 | 15.00 | 15.00 | BUY<br>estimate |
| A28 | **Electronics consumables**<br>Logic-level MOSFETs, resistor kit, perfboard, 26–30 AWG silicone wire, heat-shrink, breadboard, jumpers.<br>Ref. TC-03, EE lane. | 1 | 35.00 | 35.00 | BUY<br>estimate |
| | **Subtotal** | | | **110.00** | |

TABLE 06 Fabrication and mock-ups. 5 lines, subtotal US$205.00.

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| A29 | **PLA filament**<br>1.75 mm, 1 kg.<br>Ref. TC-01 bodies, TC-02 dummies, rig adapters. Bambu Lab PLA Basic seen at USD 19.99–22.99 in regional stores (search, 9 Oct). | 3 | 23.00 | 69.00 | BUY<br>estimate |
| A30 | **PETG filament**<br>1.75 mm, 1 kg.<br>Ref. Containers, fixtures. | 2 | 23.00 | 46.00 | BUY<br>estimate |
| A31 | **TPU 95A filament**<br>1.75 mm, 1 kg.<br>Ref. Bumpers, liner trials. | 1 | 40.00 | 40.00 | BUY<br>estimate |
| A32 | **Hatch-latch candidates**<br>Spring latches, R-clips and cotter pins, toggle latches, screw latches.<br>Ref. TC-04, SR-04, SR-15. Magnets are not an acceptable latching method (§4.2.1). | 1 | 25.00 | 25.00 | BUY<br>estimate |
| A33 | **Mock-up materials**<br>Cardboard sheets, hot-glue sticks, tape, paint markers (TOP / O / E markings, roll stripes).<br>Ref. TC-01, TC-04. | 1 | 25.00 | 25.00 | BUY<br>estimate |
| | **Subtotal** | | | **205.00** | |

## 6.0 TIER B: FORMAL UNIVERSITY REQUEST

Tier B buy-now lines total US$3,713.20, and budget holds total US$3,850.96.

TABLE 07 Radio (ExpressLRS). 6 lines, subtotal US$453.92.

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| B01 | **ELRS transmitter**<br>RadioMaster Boxer, internal ELRS 2.4 GHz, FCC version, EdgeTX. Skip if the pilot already owns a radio with internal ELRS 2.4 GHz.<br>Ref. §3.2.1, §4.2.5. Transmitter cells are not included (B02). | 1 | 153.99 | 153.99 | BUY<br>listing |
| B02 | **Transmitter cells**<br>2 × 18650 Li-ion cells and a charger, for B01.<br>Ref. B01. | 1 | 20.00 | 20.00 | BUY<br>estimate |
| B03 | **ELRS PWM receiver, 8 ch**<br>RadioMaster ER8: 8 PWM outputs, 4.5–8.4 V, ELRS 3.3.0, FCC firmware. PWM output only: no CRSF-only receivers, which need a flight controller.<br>Ref. §3.2.1, §4.2.1, §4.2.5. One per airframe plus a spare. Fail-safe positions must be set in the receiver WebUI (default is centred). | 3 | 34.99 | 104.97 | BUY<br>listing |
| B04 | **ELRS PWM receiver, 6 ch**<br>RadioMaster ER6: bench receiver for sensor and reel development, or the observer's payload link (§3.1).<br>Ref. EE lane, §3.1. | 1 | 24.99 | 24.99 | BUY<br>listing |
| B05 | **Receiver and servo battery**<br>2S LiFe 6.6 V (7.2 V full), about 1450 mAh, COTS with label.<br>Ref. §3.2.3. A separate receiver battery is mandatory. 7.2 V full charge needs HV servos (B19). | 3 | 34.99 | 104.97 | BUY<br>listing |
| B06 | **External receiver switch**<br>Heavy-duty switch harness, mounted outside the airplane. An arming plug does not count as this switch.<br>Ref. §3.2.1. | 3 | 15.00 | 45.00 | BUY<br>estimate |
| | **Subtotal** | | | **453.92** | |

TABLE 08 Propulsion and power. 17 lines, subtotal US$4,785.79.

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| B07 | **Motors: bench pair (both D9 candidates)**<br>T-Motor AT7215 30CC fixed-wing motor: 1 x KV270 (8S candidate) and 1 x KV200 (12S candidate), the two leads of the propulsion study (analysis/out/prop_trade_report.txt). Both are bench-tested (TC-06) before D9; the flight motors are B07b.<br>Ref. §3.2.1, TC-06, DB-01 D9. Unmodified commercial motor (§3.2.1). KV270: 606 g with wires (T-Motor spec table). | 2 | 219.00 | 438.00 | BUY<br>listing |
| B07b | **Motors: flight and spare (after D9)**<br>The winding chosen at D9 (KV270 for 8S or KV200 for 12S): one flight motor and one spare.<br>Ref. §3.2.1, DB-01 D9. Unmodified commercial motor (§3.2.1). | 2 | 219.00 | 438.00 | HOLD<br>listing |
| B08 | **ESC, fits both candidates**<br>Castle Creations Phoenix Edge HV 120 (010-0104-00): 3S–12S LiPo, 120 A continuous at full throttle, no internal BEC, data logging (current, voltage, RPM); 148.8 g with wires. Bench, flight and spare.<br>Ref. §3.2.1, §4.2.5, TC-06, flight power logs (evidence E04). Works on 8S and 12S, so buying it does not pre-empt D9. No BEC: the receiver runs on its own pack (B05). Programming and log download need a Castle Link USB adapter (not listed here). | 3 | 289.95 | 869.85 | BUY<br>listing |

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| B09 | **Propellers: bench sweep and spares**<br>APC Thin Electric, from the propulsion study: 6 x 20x15E (lead), 2 x 20x13E, 2 x 20.5x14E, all US$18.20; the 21x13E is B09b.<br>Ref. §3.2.1, TC-06. Commercial propellers only; tips may be clipped (§3.2.1). APC limit for E props: 150,000 / D(in) rpm, 7,500 rpm at 20 in. | 10 | 18.20 | 182.00 | BUY<br>listing |
| B09b | **Propellers: 21x13E (bench sweep)**<br>APC 21x13E Thin Electric.<br>Ref. §3.2.1, TC-06. Commercial propellers only (§3.2.1). | 2 | 21.09 | 42.18 | BUY<br>listing |
| B10 | **Propulsion pack, 8S bench candidate**<br>Thunder Power RC TP3300-8SE55: 8S 3300 mAh, Elite 55C, 97.7 Wh; 674 g (maker's estimate). Price without connector.<br>Ref. §3.2.3, §4.2.3, TC-06. Label must show chemistry, capacity, voltage and C-rating. Keep the original packaging (§4.2.3). Lithium packs ship as dangerous goods: buy locally if possible. | 1 | 151.99 | 151.99 | BUY<br>listing |
| B10b | **Propulsion pack, 12S bench candidate**<br>MaxAmps LiPo 2000 12S 44.4 V: 88.8 Wh; maximum continuous 54 A, so its fuse is capped at 54 A (§3.2.3.2); 586 g.<br>Ref. §3.2.3, §4.2.3, TC-06. Label must show chemistry, capacity, voltage and C-rating. Keep the original packaging (§4.2.3). Lithium packs ship as dangerous goods: buy locally if possible. | 1 | 372.00 | 372.00 | BUY<br>listing |
| B10c | **Propulsion packs, flight set (after D9)**<br>Four packs of the system chosen at D9: TP3300-8SE55 (US$151.99) or MaxAmps 12S 2000 (US$372.00). Held at the higher price.<br>Ref. §3.2.3, §4.2.3, DB-01 D9. Label must show chemistry, capacity, voltage and C-rating. Keep the original packaging (§4.2.3). Lithium packs ship as dangerous goods: buy locally if possible. | 4 | 372.00 | 1,488.00 | HOLD<br>listing |
| B11 | **LiPo charging sacks**<br>Commercial, unaltered, manufacturer-labelled; sized for the packs. Every battery (propulsion, receiver, sensor) is stored and charged in one.<br>Ref. §3.2.3. | 6 | 11.98 | 71.88 | BUY<br>listing |
| B12 | **Balance charger, 1–12S**<br>iCharger X12: up to 12S, 0.1–30 A, 1100 W charge power, 2 A balancing, DC input 11–53 V (needs B12b). Balance leads or adapters for the 8S and 12S packs are separate.<br>Ref. Battery care, TC-06. Replaces the 1–6S charger of 9 Oct, which cannot charge either candidate pack. | 1 | 179.99 | 179.99 | BUY<br>listing |
| B12b | **Charger power supply, 48 V**<br>Mean Well RSP-1000-48: 48 V DC, 21 A, 1008 W, AC input 90–264 V. Powers B12 at full power.<br>Ref. Battery care, TC-06. Enclosed unit with screw terminals: the AC side must be wired by someone qualified, with an earth connection. | 1 | 298.21 | 298.21 | BUY<br>listing |
| B13 | **MAXI blade fuses**<br>Assortment of MAXI blade fuses (rules Figure 5 shows a 100 A MAXI): 40, 50, 60, 80 and 100 A. Voltage rating to be confirmed before ordering: see notes.<br>Ref. §3.2.3.1, §3.2.3.2. Fuse rating ≤ the pack's maximum continuous current, capped at 100 A (§3.2.3.2); the 12S candidate pack is rated 54 A. UNVERIFIED (16 Oct): search results say the 100 A MAXI is a 32 V part and Littelfuse's 58 V MAXI series stops at 80 A; 8S is 33.6 V and 12S is 50.4 V when full. littelfuse.com blocks scripted reads: check the Littelfuse pages in a browser before ordering. | 15 | 3.00 | 45.00 | BUY<br>estimate |
| B14 | **Inline MAXI fuse holders**<br>Rated at least as high as the fuse; for 10–12 AWG. Reference: GEP waterproof MAXI kit (up to 100 A, 12–10 AWG terminals, fuse not included).<br>Ref. §3.2.3.1. One fuse harness per propulsion pack. | 6 | 16.95 | 101.70 | BUY<br>listing |

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| B15 | **Arming-plug connectors**<br>XT90-S anti-spark pairs (or the packs' connector) and XT90E-M panel mounts.<br>Ref. §3.2.2. Arming-plug wire must be the same gauge as the pack leads or heavier (10 AWG on the reference pack). | 10 | 2.50 | 25.00 | BUY<br>estimate |
| B16 | **Silicone wire**<br>10 AWG and 12 AWG, red and black, 3 m of each.<br>Ref. §3.2.2. | 1 | 30.00 | 30.00 | BUY<br>estimate |
| B17 | **Watt meter**<br>ToolkitRC WM150, 1–50 V, 150 A.<br>Ref. Thrust and energy tests. Rated 1–50 V: a full 12S pack (50.4 V) exceeds it. On the bench the thrust stand (B18) measures voltage and current; keep this meter for 8S and field checks. | 1 | 51.99 | 51.99 | BUY<br>listing |
| B18 | **Thrust stand**<br>Tyto Robotics Flight Stand 15: thrust 15 kgf, torque 8 N·m, 30,000 RPM, voltage 180 V, current 150 A (product page, 16 Oct 2026). Measures thrust, torque, RPM, voltage and current, which validating the motor maps and APC tables needs. Both candidates make about 10 kgf static and 3.4 N·m.<br>Ref. TC-06, evidence E04, proposal Table 10. QUOTE NEEDED: Tyto publishes no price (its FAQ: quotation by configuration; sales@tytorobotics.com). US$0 here is not a price. The 9 Oct entry (5 kg load-cell kit) could not hold about 100 N. Tyto also sells WindShapers (open-jet fans) for thrust at airspeed. | 1 | 0.00 | 0.00 | HOLD<br>estimate |
| | **Subtotal** | | | **4,785.79** | |

TABLE 09 Airframe and hardware. 9 lines, subtotal US$1,849.96.

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| B19 | **HV digital servos**<br>6.0–8.4 V, metal gear; size and torque from sizing.<br>Ref. §3.2.1. HV rating needed for the 2S LiFe receiver pack (7.2 V full). | 16 | 45.00 | 720.00 | HOLD<br>estimate |
| B20 | **Linkage hardware**<br>Clevises with locking sleeves, commercial control horns, pushrods.<br>Ref. §3.2.1, §4.2.1. Clevises need a mechanical lock. Horns go on per the manufacturer's instructions, never glued to film alone. | 1 | 40.00 | 40.00 | BUY<br>estimate |
| B21 | **Fasteners and threadlocker**<br>Nylon-insert lock nuts, M2–M5 screws, washers, heat-set inserts, Loctite 243.<br>Ref. §3.2.1, §4.2.1. Every permanent fastener needs a secondary retention method. | 1 | 60.00 | 60.00 | BUY<br>estimate |
| B22 | **Carbon-fibre tubes**<br>Roll-wrapped; OD and wall from sizing (spar, tail boom). Reference: Rock West 45002, 3k twill, 0.983 × 1.055 × 72 in.<br>Ref. Sizing. | 4 | 99.99 | 399.96 | HOLD<br>listing |
| B23 | **Balsa and aircraft plywood**<br>Balsa sheet 1.5–6 mm; birch aircraft ply 1.5 and 3 mm.<br>Ref. Sizing. | 1 | 150.00 | 150.00 | HOLD<br>estimate |
| B24 | **Foam boards**<br>XPS or EPS, 50 mm, for hot-wire wing cores.<br>Ref. Sizing. | 6 | 15.00 | 90.00 | HOLD<br>estimate |
| B25 | **Covering**<br>Heat-shrink covering film or laminating film (the team has no composites capability).<br>Ref. Manufacturing plan. | 4 | 30.00 | 120.00 | HOLD<br>estimate |

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| B26 | **Landing gear and wheels**<br>Set by the gear trade (tricycle or taildragger).<br>Ref. Configuration trade. | 2 | 60.00 | 120.00 | HOLD<br>estimate |
| B27 | **Adhesives and shop consumables**<br>CA (thin, medium, thick) and accelerator, 5- and 30-minute epoxy, hot glue, sandpaper, blades, zip ties, hook-and-loop, heat-shrink.<br>Ref. Build. The 2026 judges flagged missing consumables in the budget. | 1 | 150.00 | 150.00 | BUY<br>estimate |
| | **Subtotal** | | | **1,849.96** | |

TABLE 10 Sensor, D&R and test box. 7 lines, subtotal US$324.49.

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| B28 | **Slip rings**<br>Capsule slip ring, 22 mm flange, 6 wires, 2 A, up to 300 rpm (Adafruit 736).<br>Ref. SR-06, SR-07. | 3 | 14.95 | 44.85 | BUY<br>listing |
| B29 | **Reel actuator candidates**<br>Multi-turn sail-winch servo or DC gearmotor with encoder; torque and turns from Script 5.<br>Ref. SR-06, Script 5. | 3 | 45.00 | 135.00 | HOLD<br>estimate |
| B30 | **Tow-line conductor**<br>2-conductor 28–30 AWG PTFE or silicone wire, 10 m. Strength member: the PE braid (A11).<br>Ref. SR-06, SR-07. The tow line carries both the mechanical and the electrical connection (§3.1.1). | 2 | 12.00 | 24.00 | BUY<br>estimate |
| B31 | **Locking micro connectors**<br>JST-GH 2–4 pin with pre-crimped leads, for the tow-line electrical interface.<br>Ref. SR-07. | 1 | 20.00 | 20.00 | BUY<br>estimate |
| B32 | **Sensor light-controller MCU**<br>Seeed XIAO RP2040 (21 × 17.5 mm).<br>Ref. SR-07, §3.1.1. Describe it only as payload light control. | 4 | 5.16<br>GBP 3.90 | 20.64 | BUY<br>listing |
| B33 | **Sensor test-box parts**<br>3-position switch, mating tow-line connector, printed enclosure, battery holder. Generates the same 1000 / 1500 / 2000 µs PWM as the receiver channel.<br>Ref. §3.1.1, §3.3.4, §4.1. Mandatory at tech inspection and in the Ground Mission. | 2 | 20.00 | 40.00 | BUY<br>estimate |
| B34 | **Final sensor LEDs and batteries**<br>Chosen from the TC-03 results.<br>Ref. TC-03. | 1 | 40.00 | 40.00 | HOLD<br>estimate |
| | **Subtotal** | | | **324.49** | |

TABLE 11 Logistics. 1 line, subtotal US$150.00.

| ID | ITEM AND SPECIFICATION | QTY | UNIT, USD | EXT., USD | STATUS |
|---|---|---|---|---|---|
| B35 | **Shipping crate materials**<br>Plywood, foam and hardware for a travel crate.<br>Ref. Travel. The 2026 judges flagged missing crating in the budget. | 1 | 150.00 | 150.00 | HOLD<br>estimate |
| | **Subtotal** | | | **150.00** | |

## 7.0 SOURCES

These are the reference listings behind every line marked *listing*, checked on 9 Oct 2026. 19 were read on the vendor page itself. 7 come from search results for that page, which can lag the live price. Prices and stock change, so get a quote before ordering.

- A03 Vane anemometer: HoldPeak HP-866B (eBay), US$26.99, checked 2026-10-09, from search results for the vendor page.
  https://www.ebay.ca/p/10014711598
- A07 Leaf blower: EGO LB7654, 765 cfm, 56 V kit (Ace Hardware), US$359.00, checked 2026-10-09, from search results for the vendor page.
  https://www.acehardware.com/departments/lawn-and-garden/outdoor-power-equipment/leaf-blowers/7026058
- A18 High-g accelerometer (optional): Adafruit, US$24.95, checked 2026-10-09, read on the vendor page.
  https://www.adafruit.com/product/4627
- A19 Colour LED stars: LEDSupply (regular price; 0 in stock on 9 Oct), US$5.71, checked 2026-10-09, from search results for the vendor page.
  https://www.ledsupply.com/leds/cree-xlamp-xp-e2-color-high-power-led-star
- A24 Microcontroller boards: Raspberry Pi (list price), US$4.00, checked 2026-10-09, read on the vendor page.
  https://www.raspberrypi.com/products/raspberry-pi-pico/
- B01 ELRS transmitter: RadioMaster, US$153.99, checked 2026-10-09, read on the vendor page.
  https://radiomasterrc.com/products/boxer-radio-controller-m2
- B03 ELRS PWM receiver, 8 ch: RadioMaster, US$34.99, checked 2026-10-09, read on the vendor page.
  https://radiomasterrc.com/products/er8-2-4ghz-elrs-pwm-receiver
- B04 ELRS PWM receiver, 6 ch: RadioMaster, US$24.99, checked 2026-10-09, read on the vendor page.
  https://radiomasterrc.com/products/er6-2-4ghz-elrs-pwm-receiver
- B05 Receiver and servo battery: Spektrum SPMB1450LFRX (Gator RC; out of stock on 9 Oct), US$34.99, checked 2026-10-09, read on the vendor page.
  https://www.gator-rc.com/products/spektrum-1450-receiver-battery
- B07 Motors: bench pair (both D9 candidates): T-Motor (product price, all windings), US$219.00, checked 2026-10-16, read on the vendor page.
  https://store.tmotor.com/product/at7215-fixed-wing-motor.html
- B07b Motors: flight and spare (after D9): T-Motor (product price, all windings), US$219.00, checked 2026-10-16, read on the vendor page.

https://store.tmotor.com/product/at7215-fixed-wing-motor.html

- B08 ESC, fits both candidates: Castle Creations, US$289.95, checked 2026-10-16, read on the vendor page.
  https://castlecreations.com/products/air-esc-phoenix-edge-120-hv-010-0104-00
- B09 Propellers: bench sweep and spares: APC Propellers (price list PROP-PRICE-LIST_202602), US$18.20, checked 2026-10-14, read on the vendor page.
  https://www.apcprop.com/wp-content/uploads/2026/02/PROP-PRICE-LIST_202602.xlsx
- B09b Propellers: 21x13E (bench sweep): APC Propellers (price list PROP-PRICE-LIST_202602), US$21.09, checked 2026-10-14, read on the vendor page.
  https://www.apcprop.com/wp-content/uploads/2026/02/PROP-PRICE-LIST_202602.xlsx
- B10 Propulsion pack, 8S bench candidate: Thunder Power RC, US$151.99, checked 2026-10-16, read on the vendor page.
  https://www.thunderpowerrc.com/products/tp3300-8se55
- B10b Propulsion pack, 12S bench candidate: MaxAmps, US$372.00, checked 2026-10-16, read on the vendor page.
  https://maxamps.com/products/lipo-2000-12s-44-4v-battery-pack
- B10c Propulsion packs, flight set (after D9): MaxAmps, US$372.00, checked 2026-10-16, read on the vendor page.
  https://maxamps.com/products/lipo-2000-12s-44-4v-battery-pack
- B11 LiPo charging sacks: Racers Edge (HobbyEtc), US$11.98, checked 2026-10-09, from search results for the vendor page.
  https://store.hobbyetc.com/parts/view/132874
- B12 Balance charger, 1–12S: ProgressiveRC (iCharger US distributor), US$179.99, checked 2026-10-16, read on the vendor page.
  https://progressiverc.com/products/icharger-x12
- B12b Charger power supply, 48 V: TRC Electronics (Mean Well distributor; tariff included), US$298.21, checked 2026-10-16, read on the vendor page.
  https://www.trcelectronics.com/View/Mean-Well/RSP-1000-48.shtml
- B14 Inline MAXI fuse holders: CE Auto Electric Supply, US$16.95, checked 2026-10-09, from search results for the vendor page.
  https://ceautoelectricsupply.com/product/gep-waterproof-maxi-fuse-holder-kit/
- B17 Watt meter: Pyrodrone, US$51.99, checked 2026-10-09, from search results for the vendor page.
  https://pyrodrone.com/products/toolkitrc-wm150-advanced-watt-power-meter
- B22 Carbon-fibre tubes: Rock West Composites, US$99.99, checked 2026-10-09, read on the vendor page.
  https://www.rockwestcomposites.com/roll-wrapped-carbon-fiber-tubes
- B28 Slip rings: Adafruit, US$14.95, checked 2026-10-09, read on the vendor page.
  https://www.adafruit.com/product/736
- B32 Sensor light-controller MCU: The Pi Hut (incl. VAT), GBP 3.90 (US$5.16), checked 2026-10-09, from search results for the vendor page.
  https://thepihut.com/collections/all/products/seeed-xiao-rp2040